

A Versatile Framework for Formula-Based Enforcement and Synthesis in Abstract Argumentation

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Motivation: Argumentation frameworks (AFs) may need to be revised when new evidence shows that some conclusions are incorrect. Existing approaches to **enforcement in abstract argumentation** mainly focus on **minimizing syntactic change** of the AF structure. How to revise an AF to satisfy new **acceptance constraints** while minimizing unwanted changes to both the structure and conclusions?

Contributions:

Unified framework for expressing semantic and syntactic constraints on AFs via weighted propositional formulas.

- **Generalization:** existing enforcement and synthesis notions captured as special cases while enabling new forms of enforcement.
- **Complexity analysis:** complexity results ranging from NP-completeness to Σ_3^P -completeness for major argumentation semantics.
- **Exact algorithm:** MaxSAT-based counterexample-guided abstraction refinement (CEGAR) approach for Σ_2^P -complete variants.
- **Empirical evaluation:** Scalability to moderately-sized AFs on a novel enforcement problem combining semantic and syntactic change.

FORMULA-BASED ENFORCEMENT

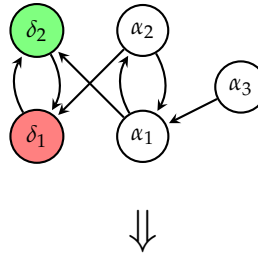
Instance

$(\underbrace{\mathcal{A}}_{\text{arguments}}, \underbrace{\Phi^+, \Phi^-, \Phi^R, w}_{\text{sets of weighted formulas}}, \underbrace{\sigma}_{\text{semantics}})$

- Φ^+ : What should happen in **at least one extension**?
- Φ^- : What must not happen in **any extension**?
- Φ^R : Constraints on the **attack structure**.
- $w(\varphi) < \infty$: soft, $w(\varphi) = \infty$: hard constraint.

Task

- Find an AF $\mathcal{F} = (\mathcal{A}, \mathcal{R})$ which
- satisfies all hard constraints, and
 - minimizes the cost incurred by not satisfying a soft constraint.

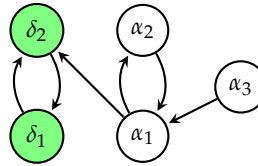


Hard constraints:

$$\Phi^+ = \underbrace{\{\text{in}_{\delta_1}, \text{in}_{\delta_2}\}}_{\delta_1 \text{ and } \delta_2 \text{ accepted}}, \Phi^- = \underbrace{\{\text{in}_{\delta_1} \wedge \text{in}_{\delta_2}\}}_{\text{not in same extension}}$$

Soft constraints:

$$\Phi^R = \{\text{att}_{a,b} \mid (a,b) \in \mathcal{R}\} \cup \underbrace{\{\neg \text{att}_{a,b} \mid (a,b) \in (\mathcal{A} \times \mathcal{A}) \setminus \mathcal{R}\}}_{\text{minimize syntactic distance}}$$



Remove attack $\alpha_2 \rightarrow \delta_1$
⇒ Stable extensions: $\{\delta_1, \alpha_2, \alpha_3\}, \{\delta_2, \alpha_2, \alpha_3\}$

COMPLEXITY RESULTS

Formula-based enforcement generalizes existing notions of extension enforcement and status enforcement.

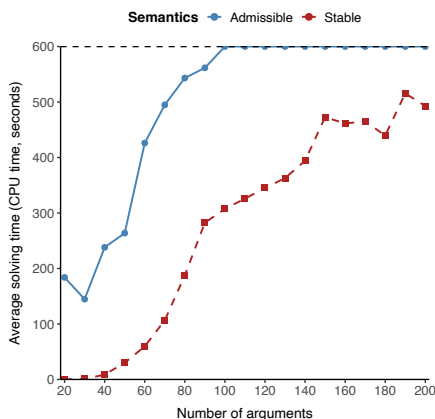
→ Substantial computational complexity.

Semantics	Complexity
Grounded	NP-complete
Admissible	Σ_2^P -complete
Complete	Σ_2^P -complete
Stable	Σ_2^P -complete
Preferred	Σ_3^P -complete

EMPIRICAL EVALUATION

Implementation available online in open source:

<https://bitbucket.org/andreasniskanen/phiaf>

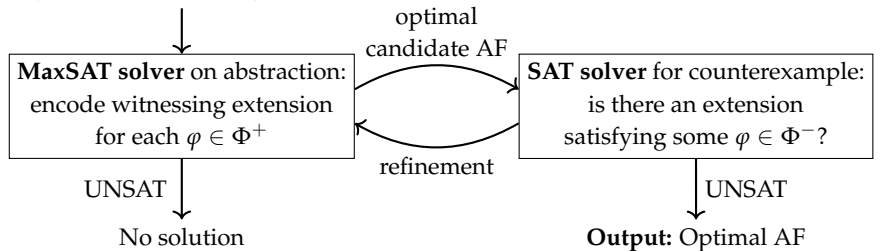


MAXSAT-BASED CEGAR ALGORITHM

Focus: Σ_2^P -complete variants (admissible, complete, and stable semantics).

- Solve **abstraction** as a MaxSAT instance:
 - encode a witnessing σ -extension for each $\varphi \in \Phi^+$ as propositional clauses
 - hard clauses: enforce each $\varphi \in \Phi^+ \cup \Phi^R$ with $w(\varphi) = \infty$
 - soft clauses: minimize cost over unsatisfied $\varphi \in \Phi^+ \cup \Phi^R$ with $w(\varphi) < \infty$
- Obtain optimal candidate AF and check for a **counterexample**:
 - SAT solver call: is there an extension satisfying some $\varphi \in \Phi^-$?
- If counterexample found: expand the abstraction by adding a **refinement clause**.
 - Effectively reducing the search space requires **strong refinements**: account for changes in the structure which preserve the counterexample extension.

Input: Enforcement instance
 $(\mathcal{A}, \Phi^+, \Phi^-, \Phi^R, w, \sigma)$



REFERENCES

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